Mathematical Methods for International Commerce

Week 10/1: Comparative Statics (5.3)

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Happy May, everyone! 🌸

I hope you had a pleasant weekend and are enjoying the extra rest.

Here is some fun economic math to make it even better.

Before we start...

Midterm exam scores are now available on the LMS.

Summary statistics:

Metric	Value
Total points	100
Average	77.0
Standard deviation	20.78
Maximum	99
Median	86.5
Minimum	23

Grades have already been uploaded to the SAINT system. If you have any questions about your score, please contact me by email.

Why It Matters in Economics, Business & Finance

> Comparative statics analyzes how equilibrium values change. It is an essential technique used to understand how changes in **policy, shocks, or parameters** affect outcomes in economic models.

- In **macroeconomics**, we can predict how national income responds to a change in government spending or taxation.
- In **finance**, understanding multipliers helps estimate how economic indicators respond to policy adjustments.
- In **microeconomics**, it shows how supply and demand curves shift in response to taxes, subsidies, or preferences.

It allows decision-makers to simulate "what if" scenarios and make **informed, strategic choices**.

Learning Objectives

By the end of this class, you should be able to:

- Derive the **reduced form** of macroeconomic models
- Compute **national income multipliers**
- Interpret **qualitatively** how changes in parameters affect outcomes
- Use **quantitative** multiplier analysis
- Apply **multipliers** in a simple one-good market model
- Distinguish between **endogenous** vs **exogenous** variables

Agenda

1. Comparative Statics (5.3)
2. Individual Assignment

1. Comparative Statics (5.3)

Structural Form → Reduced Form

Example:

Suppose:

$$Y = C + I + G$$

$$C = a + b(Y - T)$$

This is the **structural form**.

We substitute C into the Y equation:

$$Y = a + b(Y - T) + I + G$$

Solve for Y → this is the **reduced form**.

Solving for Y

Start with:

$$Y = a + b(Y - T) + I + G$$

Distribute:

$$Y = a + bY - bT + I + G$$

Group terms:

$$Y - bY = a - bT + I + G$$

Factor:

$$Y(1 - b) = a - bT + I + G$$

Solve:

$$Y = \frac{1}{1 - b}(a - bT + I + G)$$

This is the **reduced form**: it shows Y as a function of **exogenous variables**

Endogenous vs Exogenous Variables

- **Endogenous variables** are determined *inside* the model.
  - In our income model, Y (and C in structural form) is endogenous.
  - In a supply-demand model, equilibrium P^* and Q^* are endogenous.
- **Exogenous variables** are determined *outside* the model and treated as given.
  - In our reduced form, a, b, T, I, G are exogenous inputs.
  - Economic examples: fiscal policy (G, tax rate T), world interest rates, technology shocks, and weather shocks in agriculture.

If an exogenous variable changes, how do endogenous outcomes respond? For example, if G increases, how much does Y increase?

Multiplier Concept

From the reduced form:

$$Y = \frac{1}{1-b}(a - bT + I + G)$$

- The **multiplier** is $\frac{1}{1-b}$
- It shows how **sensitive** Y is to a change in **G**, **I**, etc.

If $b = 0.8$, multiplier = $\frac{1}{1-0.8} = 5$

So:

- $\Delta G = 10 \rightarrow \Delta Y = 5 \times 10 = 50$

From an economic perspective, this means that a 1-unit increase in G increases Y by 5 units.

Practice Problem

Given:

- $a = 20, b = 0.75$
- $T = 10, I = 30, G = 40$

1. Write reduced form for Y

2. Compute multiplier

3. Calculate Y

4. If G increases to 50, what is the new Y?

Solution

1. Reduced form:

$$Y = \frac{1}{1-0.75}(20 - 0.75 \cdot 10 + 30 + 40) = 4(20 - 7.5 + 30 + 40) = 4(82.5)$$

2. Multiplier = $\frac{1}{1-0.75} = 4$

3. $Y = 330$

4. New $G = 50 \rightarrow$

$$Y = 4(20 - 7.5 + 30 + 50) = 4(92.5) = 370$$

Market Model Example

Supply and demand:

$$Q_d = 100 - 5P \quad Q_s = 20 + 3P$$

Equilibrium: $Q_d = Q_s$

Solve: $100 - 5P = 20 + 3P \Rightarrow 80 = 8P \Rightarrow P^* = 10 \Rightarrow Q^* = 100 - 5 \cdot 10 = 50$

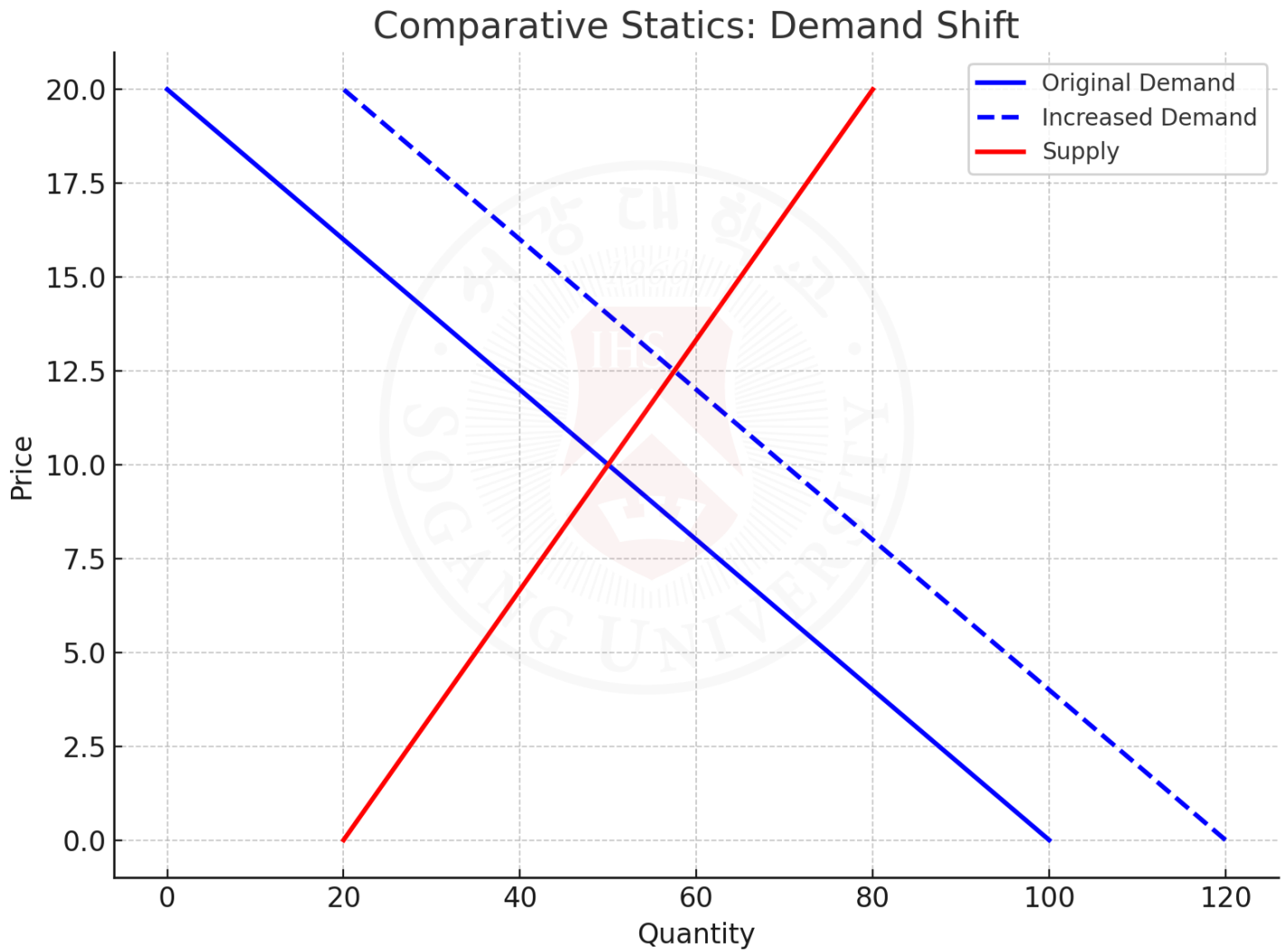
Now increase demand:

$$Q_d = 120 - 5P$$

$\rightarrow$ Solve again:
$120 - 5P = 20 + 3P \Rightarrow 100 = 8P \Rightarrow P^* = 12.5, Q^* = 120 - 5 \cdot 12.5 = 57.5$

This illustrates how a change in **parameters** shifts equilibrium. Here, the parameter shift is an intercept increase in demand (a demand shock).

Plot: Comparative Statics in Supply & Demand



Summary

- Comparative statics helps **evaluate how equilibrium changes** with a shift in exogenous variables
- Multiplier tells us the **magnitude of change**
- We derived **reduced form** from structure
- We applied the concept to:
  - National income
  - Market models

2. Individual Assignment: Comparative Statics Lab

Individual Assignment: Comparative Statics Lab

Instructions

- Work individually.
- Show all steps.
- Provide an economic interpretation for each result.
- You may use a calculator.

Task 1: Reduced Form and Multiplier

Given:

$$Y = C + I + G, \quad C = 15 + 0.8(Y - T)$$

Parameters:

$$T = 10, \quad I = 25, \quad G = 30$$

Questions:

- Derive the reduced form.
- Compute the multiplier.
- Find equilibrium Y.
- If G increases by 10, find the new Y.

Task 2: Sensitivity Analysis

Using:

$$Y = \frac{1}{1-b}(a - bT + I + G)$$

Compare:

- Case A: $b = 0.6$
- Case B: $b = 0.9$

Questions:

- Compute the multipliers.
- Which economy reacts more to policy?
- Explain why.

Task 3: Market Comparative Statics

Initial market:

$$Q_d = 100 - 4P, \quad Q_s = 20 + 2P$$

Questions:

- Find equilibrium P^* and Q^*.
- After a demand shock,

$$Q_d = 120 - 4P$$

  find the new equilibrium.
- Explain what changed.
- Explain why both P and Q increased.

Task 4: Interpretation Challenge

Short answers:

- Why does a higher b increase the multiplier?
- Is a large multiplier always good? Why or why not?

Next Classes

- (May 7) Unconstrained Optimization (5.4)

Any QUESTIONS?

Thank you for your attention and participation!